F Fluidic

Continuous-wave state synthesis

A blockless execution layer that replaces blocks with a continuously synthesized wave-field.

RAISING	VALUATION CAP	STAGE
$1.5M Seed	$17M	Testnet Live

The Problem

Blockchains are batch processors trapped in a leader-driven design.

01

Unpredictable finality

Users wait 12 seconds on Ethereum, multiple confirmations on L1s, or rely on centralized sequencers on L2s. Settlement is never truly continuous.

02

MEV and front-running

Mempools create a public auction for ordering. Searchers extract billions in value from ordinary users through sandwich attacks and frontrunning.

03

Congestion pricing

When demand spikes, fees spike. Applications become unusable for retail users, and throughput collapses under load.

04

Execution fragmentation

The Solution

Fluidic replaces blocks with continuous state synthesis.

Fluidic is a blockless Layer-1 that ingests signed state transitions called **shifts**, orders them causally, and synthesizes global state in periodic ticks.

- **No blocks.** State advances continuously through synthesis ticks.
- **No mempool auction.** Shifts are gossiped and causally ordered, not front-run.
- **Parallel by default.** Commutative shifts merge in NTT windows; stateful shifts merge through a vector-clock DAG.
- **Permissionless.** Anyone can run a synthesis node and earn rewards.
- **EVM-compatible.** Raw Ethereum transactions execute inside a revm sandbox.

Shifts Signed state transitions → **Order** NTT + vector-clock DAG → **Synthesize** Wave-field update → **Certify** BFT certificate

<1s finality, every tick

Each tick produces a signed Synthesis Certificate with Merkle roots of commutative state, stateful DAG, balances, stake, rewards, and EVM transactions.

Shifts, domains, and the wave-field.

S

Stateful Shifts

Account-to-account transfers, swaps, and EVM calls. Each carries a vector clock and predecessor hashes, forming a causal DAG. Ordered topologically and finalized after K ticks.

C

Commutative Shifts

Pool deltas, order-book net flows, and aggregated oracle updates. Order-independent by design; batched through Number-Theoretic Transform (NTT) folding for massive parallelism.

D

Concurrency Domains

Each domain is an isolated state snapshot with its own accounts, pools, NTT spectrum, and decay policy. Domains can be causal, strict, or commutative — registered permissionlessly.

Incoming Shifts
Stateful + Commutative

→

Domain Router
Per-domain policy

→

Wave-Field Update
Accounts + Pools + Spectrum

→

Synthesis Certificate
Signed by operators

Architecture

Three execution paths. One unified state.

1. Commutative Path

Order-independent deltas are accumulated and verified via NTT round-trip. Perfect for AMM pool updates, order-book net flows, and high-frequency telemetry.

- 100K+ NTT ops/sec
- No causal ordering overhead
- Deterministic batch verification

2. Stateful Path

State-dependent operations are ordered by vector-clock happened-before relationships. Double-spends and causal cycles are rejected with signed proofs.

- Vector-clock DAG
- Topological application
- Verifiable rejection proofs

3. EVM Path

Raw signed Ethereum transactions execute in a revm sandbox. EVM state is synthesized alongside native shifts, giving Solidity developers a drop-in runtime.

- revm execution engine
- Compatible with existing wallets
- Merged into synthesis certificates

Stake-weighted BFT certificates, not leader election.

How finality works

- **1. Operators synthesize.** Each tick, staked operators run the same deterministic synthesis and compute Merkle roots.
- **2. Certificates are signed.** Operators broadcast signed Synthesis Certificates containing the roots.
- **3. Quorum is reached.** A tick is finalized when **2/3 + 1** of stake-weighted operators have signed matching roots.
- **4. Equivocation is slashed.** Conflicting certificates for the same tick trigger slashing and burn the offender's stake.

2/3+1

Stake-weighted quorum for BFT finality

<1s

Default synthesis interval, configurable per domain

∞

Permissionless operators; no whitelisted sequencer set

Tokenomics

WAVE: native fuel, stake, and burn-reward loop.

Metabolic burn engine

Every synthesis tick applies continuous exponential decay to idle WAVE balances. The decayed value is split:

25%
Permanently burned

75%
Rewards to stakers + LPs

Activity grace windows protect active accounts, aligning network usage with economic participation.

Supply design

- Fixed long-term supply cap modeled around 1B WAVE
- Genesis seeds operators, faucet, and ecosystem pools
- New issuance only through operator rewards and faucet drips
- Deflationary sink grows with economic activity

Stake & rewards

- Minimum stake threshold to sign certificates
- Rewards distributed proportional to stake every tick
- Slashing for equivocation and downtime

Seedless, permissionless peer discovery.

Fluidic nodes find each other without hardcoded seed lists. Discovery layers include signed genesis DNS records, the public BitTorrent Mainline DHT, and LAN mDNS.

DNS

Signed Genesis DNS

Bootstrap records served as HTTPS JSON or DNS TXT, signed by genesis keys. Authenticated first contact.

DHT

Mainline DHT

Nodes announce themselves on a network-specific infohash derived from FLUIDIC_NETWORK_ID. Decentralized and censorship-resistant.

mDNS

LAN mDNS

Local mesh discovery over multicast DNS for development, homelab, and private datacenter deployments.

TCP

Gossip Mesh

Authenticated TCP gossip with optional PSK. Shifts, certificates, and peer announcements propagate across the mesh.

New Node
Boots with OSCILLATOR_ID

→

DNS / DHT / mDNS
Discover peers

→

State Sync
Download snapshot

→

Join Mesh
Synthesize & certify

Product Stack

Everything needed to run, build, and use the network.

N

Fluidic Node

Rust runtime packaged as a single Docker image. Runs as a light client by default; becomes a full operator when a PUBLIC_ENDPOINT is advertised. Includes persistence, API, and gossip.

S

TypeScript SDK

Generate keys, register accounts, sign stateful and commutative shifts, query balances, and subscribe to WebSocket feeds. Drop-in client for wallets and dApps.

D

Reference dApp

Live example at app.testnet.fluidic.foundation: wallet creation, faucet, swaps, and transaction history. Open-source and forkable.

E

Explorer

Blockless explorer showing shifts, ticks, certificates, validators, and quorum status. Integrated into the landing page.

F

Testnet Faucet

Distributes WAVE and USDC test tokens to registered accounts. Containerized and deployed alongside the testnet mesh.

B

EVM Bridge Sandbox

Execute raw Ethereum transactions inside the synthesis loop. Supports existing wallets and Solidity contracts without rewrites.

Traction

Testnet is live. Code is shipping.

Live

Testnet mesh running on Railway with public API endpoints

81+

Unit and integration tests passing in CI

100K+

Commutative NTT ops/sec in benchmarks

<1s

Synthesis-tick finality on public testnet

Milestone	Status	Details
Core runtime	✓ Shipped	Wave-field, NTT engine, vector-clock DAG, metabolic decay, EVM sandbox
Networking	✓ Shipped	TCP gossip, Mainline DHT, mDNS, signed DNS/HTTPS bootstrap records
Consensus	✓ Shipped	Stake table, BFT quorum certificates, slashing, light client
Developer stack	✓ Shipped	REST + WebSocket API, TypeScript SDK, reference dApp, explorer, docs
Public testnet	✓ Live	api.testnet.fluidic.foundation + Railway-orchestrated services

Market

Built for the applications that break blockchains.

D

DeFi

CLOBs and AMMs on one engine. MEV-resistant causal ordering. Cross-domain composability without bridge latency.

A

AI Agents

Per-agent state domains. Verifiable action logs. Pay-per-signal economics. Agents can transact at machine speed without fee spikes.

G

Gaming

Tick-synchronized game state. High-frequency player actions. Asset ownership without block congestion.

I

Institutions

Auditable Merkle certificates. Private compliance domains with strict ordering. Deterministic settlement windows.

> The total addressable market is the intersection of high-frequency decentralized finance, autonomous agent economies, and real-time on-chain applications — all of which need sub-second, fair-ordered, parallelizable state.

Roadmap

From testnet to production mesh.

Now — Q2 2026

Testnet Hardening

- Public testnet stability
- SDK + dApp developer onboarding
- Security audits (runtime, consensus, economics)
- Seedless discovery at scale
- Bug bounty program

Q3 — Q4 2026

Mainnet Readiness

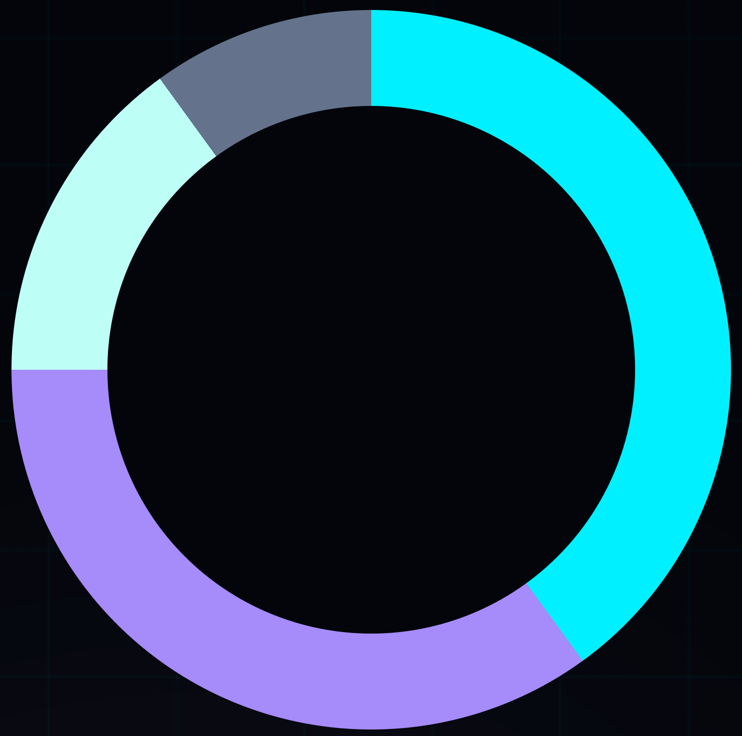
- Incentivized testnet with real rewards
- Bridge adapters for Ethereum, Solana, Bitcoin
- Domain marketplace and policy tooling
- Validator set expansion
- Mainnet genesis design finalization

2027+

Production Mesh

- Mainnet launch with WAVE genesis
- Institutional compliance domains
- AI-agent runtime partnerships
- Cross-chain liquidity hubs
- Decentralized governance upgrade path

$1.5M seed at a $17M valuation cap.



40% — Engineering & Protocol
Core runtime hardening, consensus audits, performance optimization, and developer tooling.

35% — Security & Audits
Third-party audits, formal verification, fuzzing, bug bounties, and testnet attack campaigns.

15% — Ecosystem & BD
Grants, integrations, reference applications, and strategic partnerships.

10% — Operations
Legal, infrastructure, testnet operations, and administrative runway.

Fluidic

Join the blockless state synthesis mesh.

We are replacing the block with a wave. If you believe decentralized state should advance continuously, fairly, and in parallel, let's build the next execution layer together.

Run a node
ghcr.io/fluidic-foundation/fluidic-fvm/mesh-node:latest

Explore the testnet
https://testnet.fluidic.foundation/explorer

Read the source
https://github.com/Fluidic-Foundation/Fluidic-FVM

RAISING
$1.5M Seed

CAP
$17M

CONTACT
hello@fluidic.foundation